

## Problem 4

## Problem 4

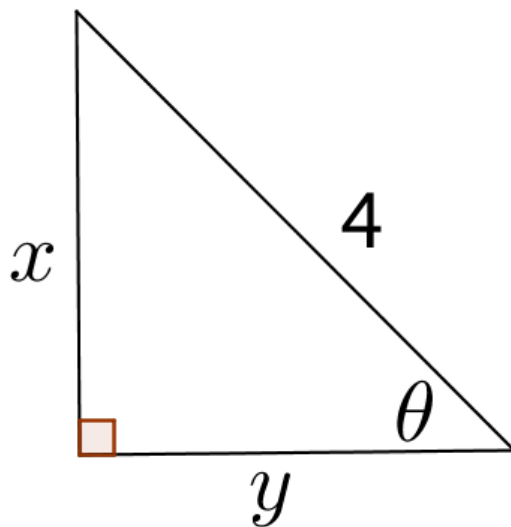
**Problem 1** Simplify each of the following expressions.

(a)  $\cos^{-1}(\sin(\pi/2))$

**Explanation.** By the unit circle,  $\sin(\pi/2) = 1$ , and so we are looking for  $\cos^{-1}(1)$ . The range of  $\cos^{-1}$  is  $[0, \pi]$ , and so, by properties of inverse functions,  $\cos^{-1}(1) = 0$ .

(b)  $\tan(\sin^{-1}(x/4))$

**Explanation.** Let  $\theta = \sin^{-1}(x/4)$ , then  $\sin(\theta) = x/4$ . We can then draw the corresponding right triangle:



Calling the adjacent side  $y$ , by the Pythagorean Theorem we obtain

$$4^2 = x^2 + y^2 \implies y = \sqrt{16 - x^2}$$

Remark: Since  $\theta$  is in the range of  $\sin^{-1}$ , it follows that  $-\pi/2 < \theta < \pi/2$ . Therefore,  $\cos(\theta) = y/4 > 0$ . Therefore,  $y > 0$ .

*Problem 4*

Then

$$\begin{aligned}\tan(\sin^{-1}(4/x)) &= \tan(\theta), \\ &= \frac{x}{y}, \\ &= \frac{x}{\sqrt{16-x^2}}.\end{aligned}$$

Note:  $\tan \theta$  has the same sign as  $x$ , since  $y > 0$ .